



LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Whole Blood EDTA	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMP	Ref Id1 :
Report Date and Time : 30-Mar-2022 09:23	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF. INTERVAL	REMARKS
------	---------	------	--------------------------	---------

HAEMOGRAM REPORT

HB AND INDICES

Haemoglobin (Colorimetric)	14.6	G%	13.00 - 17.00
RBC (Electrical Impedance)	5.29	millions/cumm	4.50 - 5.50
PCV(Calc)	42.48	%	40.00 - 50.00
MCV (RBC histogram)	L 80.3	fL	83.00 - 101.00
MCH (Calc)	27.6	pg	27.00 - 32.00
MCHC (Calc)	34.4	gm/dL	31.50 - 34.50
RDW (RBC histogram)	13.70	%	11.00 - 16.00

TOTAL AND DIFFERENTIAL WBC COUNT (Flowcytometry)

Total WBC Count	6860	/μL	4000.00 - 10000.00		
	[%]		EXPECTED VALUES	[Abs]	EXPECTED VALUES
Neutrophil	50.9	%	40.00 - 70.00	3492	/ul 2000.00 - 7000.00
Lymphocyte	38.8	%	20.00 - 40.00	2662	/ul 1000.00 - 3000.00
Eosinophil	4.6	%	1.00 - 6.00	316	/ul 20.00 - 500.00
Monocytes	5.4	%	2.00 - 10.00	370	/ul 200.00 - 1000.00
Basophil	0.3	%	0.00 - 2.00	21	/ul 0.00 - 100.00
Neutrophil to Lymphocyte Ratio (NLR)	1.31		0.78 - 3.53		

PLATELET COUNT (Optical)

Platelet Count	316000	/μL	150000.00 - 410000.00
----------------	--------	-----	-----------------------

SMEAR STUDY

RBC Morphology	Normocytic Normochromic RBCs.
WBC Morphology	Total WBC count within normal limits.
Platelet	Platelets are adequate in number.
Parasite	Malarial Parasite not seen on smear.

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr. Pavan Dave
DCP, DNB (PATH)

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 1 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Whole Blood EDTA	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMPL	Ref Id1 :
Report Date and Time : 30-Mar-2022 10:10	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
------	---------	------	----------------------	---------

HAEMATOLOGY INVESTIGATIONS

ESR Photometrical capillary stopped flow kinetic analysis	6	mm after 1hr	3 - 15	
---	---	--------------	--------	--

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 2 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Plasma Fluoride F	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMPL	Ref Id1 :
Report Date and Time : 30-Mar-2022 09:50	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
------	---------	------	----------------------	---------

BIOCHEMICAL INVESTIGATIONS

Plasma Glucose - F <i>Photometric, Hexokinase</i>	85.0	mg/dL	70.0 - 100	
---	-------------	-------	------------	--

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 3 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Whole Blood EDTA	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMPL	Ref Id1 :
Report Date and Time : 30-Mar-2022 11:33	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
BIOCHEMICAL INVESTIGATIONS				
Glycated Haemoglobin Estimation				
HbA1C HPLC	5.20	% of total Hb	<5.7: Normal 5.7-6.4: Prediabetes ≥6.5: Diabetes	
Avg. Pl Glucose (Last 3 Months) Calculated	102.54	mg/dL	80.00 - 140.00	

Please Note change in reference range as per ADA 2021 guidelines.

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 4 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : **Mr. DHRISHIT SHAH** Sex/Age : **Male / 16 Years** Case ID : **20300145570**
 Ref. By : Dis. At : Pt. ID : 215056
 Bill. Loc. : Pt. Loc. :

Reg Date and Time : 30-Mar-2022 07:16 Sample Type : Serum Mobile No. : 7819811953
 Sample Date and Time : 30-Mar-2022 07:16 Sample Coll. By : STMPL Ref Id1 :
 Report Date and Time : 30-Mar-2022 10:59 Acc. Remarks Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
------	---------	------	----------------------	---------

BIOCHEMICAL INVESTIGATIONS

Renal Function Test

Urea <i>Urease/GLDH</i>	25.00	mg/dL	18.0 - 45.0	
Creatinine <i>Kinetic Alkaline Picrate</i>	0.79	mg/dL	0.70 - 1.30	
Uric Acid <i>Uricase</i>	7.20	mg/dL	3.5 - 7.2	
Sodium <i>ISE</i>	139.00	mmol/L	136 - 145	
Potassium <i>ISE</i>	4.30	mmol/L	3.5 - 5.1	
Chloride <i>ISE</i>	103.00	mmol/L	98 - 107	

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 5 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Serum	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMP	Ref Id1 :
Report Date and Time : 30-Mar-2022 10:30	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
BIOCHEMICAL INVESTIGATIONS				
Liver Function Test				
S.G.P.T. <i>NADH (Without P-5-P)</i>	30.00	U/L	0.0 - 63	
S.G.O.T. <i>NADH (Without P-5-P)</i>	20.00	U/L	0 - 37	
Alkaline Phosphatase <i>Enzymatic, PNPP-AMP</i>	H 182.00	U/L	40 - 150	
Gamma Glutamyl Transferase <i>Multipoint Rate-L-y-Glytamyl-p-nitroanilide</i>	15.00	U/L	0 - 64	
Proteins (Total) <i>Biuret</i>	7.50	gm/dL	6.40 - 8.30	
Albumin <i>(BCG)</i>	4.50	gm/dL	3.4 - 5	
Globulin <i>CALC</i>	3.00	gm/dL	2 - 4.1	
A/G Ratio <i>CALC</i>	1.5		1 - 2.1	
Bilirubin Total <i>Colorimetric Diazo Method</i>	0.3	mg/dL	0.30 - 1.20	
Bilirubin Conjugated <i>Colorimetric Diazo Method</i>	0.12	mg/dL	0 - 0.50	
Bilirubin Unconjugated <i>CALC</i>	0.18	mg/dL	0 - 0.8	

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr. Pavan Dave
DCP, DNB (PATH)

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 6 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Serum	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMPL	Ref Id1 :
Report Date and Time : 30-Mar-2022 10:21	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
------	---------	------	----------------------	---------

BIOCHEMICAL INVESTIGATIONS

Skeletal Profile

Calcium OCPC	9.40	mg/dL	8.4 - 10.2
Phosphorus Inorganic Phosphomolybdate	5.70	mg/dL	3.5 - 5.9

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 7 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : **Mr. DHRISHIT SHAH** Sex/Age : **Male / 16 Years** Case ID : **20300145570**
 Ref. By : Dis. At : Pt. ID : **215056**
 Bill. Loc. : Pt. Loc. :

Reg Date and Time : 30-Mar-2022 07:16 Sample Type : Serum Mobile No. : 7819811953
 Sample Date and Time : 30-Mar-2022 07:16 Sample Coll. By : STMP L Ref Id1 :
 Report Date and Time : 30-Mar-2022 10:30 Acc. Remarks Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
------	---------	------	----------------------	---------

BIOCHEMICAL INVESTIGATIONS

Lipid Profile

Cholesterol <i>Enzymatic</i>	165.00	mg/dL	110 - 200	
HDL Cholesterol <i>Accelerator Selective Detergent</i>	L 29.0	mg/dL	40 - 60	
Triglyceride <i>Colorimetric-Arsenazo Method</i>	H 368.00	mg/dL	40 - 200	? fasting sample. Adv : Repeat sample after proper 10 to 12 hour of fasting or after 1 week
VLDL <i>Calculated</i>	H 73.60	mg/dL	10 - 40	
Chol/HDL <i>Calculated</i>	H 5.69		0 - 4.1	
LDL Cholesterol (Direct) <i>Enzymatic</i>	90.00	mg/dL	0 - 100	

NEW ATP III GUIDELINES (MAY 2001), MODIFICATION OF NCEP

LDL CHOLESTEROL	CHOLESTEROL	HDL CHOLESTEROL	TRIGLYCERIDES
Optimal<100	Desirable<200	Low<40	Normal<150
Near Optimal 100-129	Border Line 200-239	High >60	Border High 150-199
Borderline 130-159	High >240	-	High 200-499
High 160-189	-	-	-

- LDL Cholesterol level is primary goal for treatment and varies with risk category and assesment
- For LDL Cholesterol level Please consider direct LDL value
- Risk assesment from HDL and Triglyceride has been revised. Also LDL goals have changed.
- Detail test interpreation available from the lab
- All tests are done according to NCEP guidelines and with FDA approved kits.
- LDL Cholesterol level is primary goal for treatment and varies with risk category and assesment

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr. Pavan Dave

Dr. Pavan Dave
DCP, DNB (PATH)

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 8 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Spot Urine	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMPL	Ref Id1 :
Report Date and Time : 30-Mar-2022 09:37	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
------	---------	------	----------------------	---------

URINE EXAMINATION (STRIP METHOD AND FLOWCYTOMETRY)

Physical Examination

Colour : Pale Yellow

Transparency : Clear

Chemical Examination By Sysmex UC-3500

Sp.Gravity	1.0300	1.005 - 1.03
pH	5.5	5.0 - 8.0
Leucocytes (ESTERASE)	Negative	Negative
Protein	Negative	Negative
Glucose	Negative	Negative
Ketone Bodies Urine	Negative	Negative
Urobilinogen	Negative	Negative
Bilirubin	Negative	Negative
Blood	Negative	Negative
Nitrite	Negative	Negative

Flowcytometric Examination By Sysmex UF-5000

Leucocyte	Nil	/HPF	Nil
Red Blood Cell	Nil	/HPF	Nil
Epithelial Cell	Present(+)	/HPF	Present(+)
Bacteria	Nil	/μL	Nil
Yeast	Nil	/μL	Nil
Cast	Nil	/LPF	Nil
Crystals	Negative		Negative

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr. Pavan Dave
DCP, DNB (PATH)

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 9 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH Sex/Age : Male / 16 Years Case ID : 20300145570
 Ref. By : Dis. At : Pt. ID : 215056
 Bill. Loc. : Pt. Loc. :
 Reg Date and Time : 30-Mar-2022 07:16 Sample Type : Spot Urine Mobile No. : 7819811953
 Sample Date and Time : 30-Mar-2022 07:16 Sample Coll. By : STMPL Ref Id1 :
 Report Date and Time : 30-Mar-2022 09:37 Acc. Remarks Ref Id2 :

Parameter	Unit	Expected value	Result/Notations				
			Trace	+	++	+++	++++
pH	-	4.8-7.4					
SG	-	1.016-1.022					
Protein	mg/dL	Negative (<10)	10	25	75	150	50
Glucose	mg/dL	Negative (<30)	30	50	100	300	1000
Bilirubin	mg/dL	Negative (0.2)	0.2	1	3	6	-
Ketone	mg/dL	Negative (<5)	5	15	50	150	-
Urobilinogen	mg/dL	Negative (<1)	1	4	8	12	-

Parameter	Unit	Expected value	Result/Notations				
			Trace	+	++	+++	++++
Leukocytes (Strip)	/micro L	Negative (<10)	10	25	100	500	-
Nitrite(Strip)	-	Negative	-	-	-	-	-
Erythrocytes(Strip)	/micro L	Negative (<5)	10	25	50	150	250
Pus cells (Microscopic)	/hpf	<5	-	-	-	-	-
Red blood cells(Microscopic)	/hpf	<2	-	-	-	-	-
Cast (Microscopic)	/lpf	<2	-	-	-	-	-

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr. Pavan Dave
DCP, DNB (PATH)

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 10 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Serum	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMP	Ref Id1 :
Report Date and Time : 30-Mar-2022 10:26	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	TEST REMARK
Thyroid Function Test				
Triiodothyronine (T3) CMIA	1.11	ng/mL	0.86 - 1.92	
Thyroxine (T4) CMIA	7.7	µg/dL	4.6 - 10.5	
TSH CMIA	2.9581	µIU/mL	0.7 - 6.4	

Interpretation Note:

Ultra sensitive-thyroid-stimulating hormone (TSH) is a highly effective screening assay for thyroid disorders. In patients with an intact pituitary-thyroid axis, s-TSH provides a physiologic indicator of the functional level of thyroid hormone activity. Increased s-TSH indicates inadequate thyroid hormone, and suppressed s-TSH indicates excess thyroid hormone. Transient s-TSH abnormalities may be found in seriously ill, hospitalized patients, so this is not the ideal setting to assess thyroid function. However, even in these patients, s-TSH works better than total thyroxine (an alternative screening test). When the s-TSH result is abnormal, appropriate follow-up tests T4 & free T3 levels should be performed. If TSH is between 5.0 to 10.0 & free T4 & free T3 level are normal then it is considered as subclinical hypothyroidism which should be followed up after 4 weeks & If TSH is > 10 & free T4 & free T3 level are normal then it is considered as overt hypothyroidism.

Serum triiodothyronine (T3) levels often are depressed in sick and hospitalized patients, caused in part by the biochemical shift to the production of reverse T3. Therefore, T3 generally is not a reliable predictor of hypothyroidism. However, in a small subset of hyperthyroid patients, hyperthyroidism may be caused by overproduction of T3 (T3 toxicosis). To help diagnose and monitor this subgroup, T3 is measured on all specimens with suppressed s-TSH and normal FT4 concentrations.

Normal ranges of TSH & thyroid hormones vary according trimester in pregnancy.

TSH ref range in Pregnancy	Reference range (microlU/ml)
First trimester	0.24 - 2.00
Second trimester	0.43-2.2
Third trimester	0.8-2.5

	T3	T4	TSH
Normal Thyroid function	N	N	N
Primary Hyperthyroidism	↑	↑	↓
Secondary Hyperthyroidism	↑	↑	↑
Grave's Thyroiditis	↑	↑	↑
T3 Thyrotoxicosis	↑	N	N/↓
Primary Hypothyroidism	↓	↓	↑
Secondary Hypothyroidism	↓	↓	↓
Subclinical Hypothyroidism	N	N	↑
Patient on treatment	N	N/↑	↓

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Sandip Shah

Dr Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 11 of 13

Printed On : 30-Mar-2022 11:37



Neuberg Supratech Reference Laboratories Private Limited

"KEDAR" Opposite Krupa Petrol Pump, Near Parimal Garden, Ahmedabad - 380006 | Phone : 079-40408181 / 61618181
Email : contact@supratechlabs.com | Website : www.supratechlabs.com | CIN : U85195GJ2013PTC077365,2013-14



LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Serum	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMP	Ref Id1 :
Report Date and Time : 30-Mar-2022 10:26	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
BIOCHEMICAL INVESTIGATIONS				

25-OH Cholecalciferol CMIA	55.3	ng/mL	Adult: 30 - 100 Normal Level 20 - <30 Insufficiency < 20 Deficiency > 150 Toxicity Pediatric 20 - 100 Normal Level 15 - <20 Insufficiency < 15 Deficiency
--------------------------------------	-------------	-------	---

25-OH-VitD plays a primary role in the maintenance of calcium homeostasis. It promotes intestinal calcium absorption and, in concert with PTH, skeletal calcium deposition, or less commonly, calcium mobilization. Modest 25-OH-VitD deficiency is common; in institutionalised elderly, its prevalence may be >50%. Although much less common, severe deficiency is not rare either. Reasons for suboptimal 25-OH-VitD levels include lack of sunshine exposure, a particular problem in Northern latitudes during winter; inadequate intake; malabsorption (e.g. due to Celiac disease); depressed hepatic vitamin D 25-hydroxylase activity, secondary to advanced liver disease; and enzyme-inducing drugs, in particular many antiepileptic drugs, including phenytoin, phenobarbital, and carbamazepine, that increase 25-OH-VitD metabolism. Hypervitaminosis D is rare, and is only seen after prolonged exposure to extremely high doses of vitamin D. When it occurs, it can result in severe hypercalcemia and hyperphosphatemia.

INTERPRETATION

- Levels <10 ng/mL may be associated with more severe abnormalities and can lead to inadequate mineralization of newly formed osteoid, resulting in rickets in children and osteomalacia in adults. In these individuals, serum calcium levels may be marginally low, and parathyroid hormone (PTH) and serum alkaline phosphatase are usually elevated. Definitive diagnosis rests on the typical radiographic findings or bone biopsy/histomorphometry.
- Patients who present with hypercalcemia, hyperphosphatemia, and low PTH may suffer either from ectopic, unregulated conversion of 25-OH-VitD to 1,25 (OH)²-VitD, as can occur in granulomatous diseases, particularly sarcoidosis, or from nutritionally-induced hypervitaminosis D. Serum 1,25 (OH)²-VitD levels will be high in both groups, but only patients with hypervitaminosis D will have serum 25-OH-VitD concentrations of >80 ng/mL, typically >150 ng/mL.
- Patients with CKD have an exceptionally high rate of severe vitamin D deficiency that is further exacerbated by the reduced ability to convert 25-OH-VitD into the active form, 1,25 (OH)²-VitD. Emerging evidence also suggests that the progression of CKD & many of the cardiovascular complications may be linked to hypovitaminosis D.
- Approximately half of Stage 2 and 3 CKD patients are nutritional vitamin D deficient (25-OH-VitD, less than 30 ng/mL), and this deficiency is more common among stage 4 CKD patients. Additionally, calcitriol (1,25 (OH)²-VitD) levels are also overtly low (less than 22 pg/mL) in CKD patients. Similarly, vast majority of dialysis patients are found to be deficient in nutritional vitamin D and have low calcitriol levels. Recent data suggest an elevated PTH is a poor indicator of deficiencies of nutritional vitamin D and calcitriol in CKD patients. CAUTIONS Long term use of anticonvulsant medications may result in vitamin D deficiency that could lead to bone disease; the anticonvulsants most implicated are phenytoin, phenobarbital, carbamazepine, and valproic acid.

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Signature

Dr Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 12 of 13

Printed On : 30-Mar-2022 11:37





LABORATORY REPORT



Name : Mr. DHRISHIT SHAH	Sex/Age : Male / 16 Years	Case ID : 20300145570
Ref. By :	Dis. At :	Pt. ID : 215056
Bill. Loc. :		Pt. Loc. :
Reg Date and Time : 30-Mar-2022 07:16	Sample Type : Serum	Mobile No. : 7819811953
Sample Date and Time : 30-Mar-2022 07:16	Sample Coll. By : STMP	Ref Id1 :
Report Date and Time : 30-Mar-2022 10:26	Acc. Remarks	Ref Id2 :

TEST	RESULTS	UNIT	BIOLOGICAL REF RANGE	REMARKS
BIOCHEMICAL INVESTIGATIONS				

Vitamin B - 12 Level **304.00** pg/mL 211 - 911

CMIA

Introduction :

Vitamin B12, a member of the corrin family, is a cofactor for the formation of myelin, and along with folate, is required for DNA synthesis. Levels above 300 or 400 are rarely associated with B12 deficiency induced hematological or neurological disease.

Clinical Significance :

Causes of Vitamin B12 deficiency can be divided into three classes: Nutritional, malabsorption syndromes and gastrointestinal causes. B12 deficiency can cause Megaloblastic anemia (MA), nerve damage and degeneration of the spinal cord. Lack of B12 even mild deficiencies damages the myelin sheath. The nerve damage caused by a lack of B12 may become permanently debilitating.

The relationship between B12 and MA is not always clear that some patients with MA will have normal B12 levels; conversely, many individuals with B12 deficiency are not afflicted with MA.

Decreased in:

Iron deficiency, normal near-term pregnancy, vegetarianism, partial gastrectomy/ileal damage, celiac disease, use of oral contraception, parasitic competition, pancreatic deficiency, treated epilepsy and advancing age.

Increased in:

Renal failure, liver disease and myeloproliferative diseases.

Variations due to age Increases: with age.

Temporarily Increased after Drug.

Falsely high in Deteriorated sample.

----- End Of Report -----

For test performed on specimens received or collected from non-NSRL locations, it is presumed that the specimen belongs to the patient named or identified as labeled on the container/test request and such verification has been carried out at the point generation of the said specimen by the sender. NSRL will be responsible Only for the analytical part of test carried out. All other responsibility will be of referring Laboratory.

Note:(LL-VeryLow,L-Low,H-High,HH-VeryHigh ,A-Abnormal)

Dr Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Dr. Sandip Shah
M.D. (Path. & Bact.)
Consultant Pathologist

Page 13 of 13

Printed On : 30-Mar-2022 11:37

